

Task 13-(AIML) - Assignment Question

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Domain- AIML

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Branch- Artificial Intelligence And Machine Learning

Git Hub Repository: <https://github.com/ChaitanyaBarge/Task-13.git>

Q.1

Ans

Program Code:

```
import numpy as np
array1 = np.array([10, 20, 30, 40, 50])
array2 = np.array([
    [1, 2, 3],
    [4, 5, 6],
    [7, 8, 9]])
print("1D Array:")
print(array1)
print("Shape:", array1.shape)
print("Data Type:", array1.dtype)
print()
print("2D Array:")
print(array2)
print("Shape:", array2.shape)
print("Data Type:", array2.dtype)
```

Output:

```
PS C:\Users\Chaitanya Barge\Desktop\Git Folder> & "C:\Users\Chaitanya Barge\AppData\Local\Programs\none.exe" "c:/Users/Chaitanya Barge/Desktop/Git Folder/array.py"
1D Array:
[10 20 30 40 50]
Shape: (5,)
Data Type: int64

2D Array:
[[1 2 3]
 [4 5 6]
 [7 8 9]]
Shape: (3, 3)
Data Type: int64
```

Q.2

Ans

Program Code:

```
#Q.2
import numpy as np
array1 = np.zeros(8)
array2 = np.ones((4, 4))
array3 = np.zeros((3, 3))
print("a) 1D Array of 8 Zeros:")
print(array1)

print("\nb) 4x4 Array of Ones:")
print(array2)

print("\nc) 3x3 Matrix of Zeros:")
print(array3)
```

Output:

```

PS C:\Users\Chaitanya Barge\Desktop\Git Folder> & "C:\Users\Chaitanya Barge\AppData\Local\Programs\Python\Python38\python.exe" "C:\Users\Chaitanya Barge\Desktop\Git Folder\array.py"
a) 1D Array of 8 Zeros:
[0. 0. 0. 0. 0. 0. 0. 0.]

b) 4x4 Array of Ones:
[[1. 1. 1. 1.]
 [1. 1. 1. 1.]
 [1. 1. 1. 1.]
 [1. 1. 1. 1.]]

c) 3x3 Matrix of Zeros:
[[0. 0. 0.]
 [0. 0. 0.]
 [0. 0. 0.]]

```

Q.3

Ans

Program Code:

```

#Q.3
import numpy as np
array1 = np.arange(0, 21, 1)
print("a) Numbers from 0 to 20:")
print(array1)
array2 = np.arange(10, 51, 2)
print("\nb) Even numbers from 10 to 50:")
print(array2)
array3 = np.arange(5, 101, 5)
print("\nc) Numbers from 5 to 100 with step of 5:")
print(array3)

```

Output:

```

PS C:\Users\Chaitanya Barge\Desktop\Git Folder> & "C:\Users\Chaitanya Barge\AppData\Local\Programs\Python\Python38\python.exe" "C:\Users\Chaitanya Barge\Desktop\Git Folder\array.py"
a) Numbers from 0 to 20:
[ 0  1  2  3  4  5  6  7  8  9 10 11 12 13 14 15 16 17 18 19 20]

b) Even numbers from 10 to 50:
[10 12 14 16 18 20 22 24 26 28 30 32 34 36 38 40 42 44 46 48 50]

c) Numbers from 5 to 100 with step of 5:
[ 5 10 15 20 25 30 35 40 45 50 55 60 65 70 75 80 85 90 95 100]

```

Q.4

Ans

Program Code:

```
#Q.4
import numpy as np
array1 = np.linspace(0, 5, 10)
print("a) 10 equally spaced numbers between 0 and 5:")
print(array1)
print("Length:", len(array1))
array2 = np.linspace(-10, 10, 15)
print("\nb) 15 equally spaced numbers between -10 and 10:")
print(array2)
print("Length:", len(array2))
```

Output:

```
PS C:\Users\Chaitanya Barge\Desktop\Git Folder> & "C:\Users\Chaitanya Barge\AppData\Local\Programs\Python\Python39\python.exe" C:\Users\Chaitanya Barge\AppData\Local\Programs\Python\Python39\python.exe C:\Users\Chaitanya Barge\Desktop\Git Folder\array.py
a) 10 equally spaced numbers between 0 and 5:
[0.          0.55555556 1.11111111 1.66666667 2.22222222 2.77777778
 3.33333333 3.88888889 4.44444444 5.          ]
Length: 10

b) 15 equally spaced numbers between -10 and 10:
[-10.         -8.57142857 -7.14285714 -5.71428571 -4.28571429
 -2.85714286 -1.42857143  0.          1.42857143  2.85714286
 4.28571429  5.71428571  7.14285714  8.57142857 10.          ]
Length: 15
```

Q.5

Ans

Program Code:

```
#Q.5
import numpy as np
array1 = np.random.rand(10)
print("a) 1D array of 10 random numbers between 0 and 1:")
print(array1)
array2 = np.random.randn(3, 3)
print("\nb) 3x3 matrix of random numbers (standard normal distribution):")
print(array2)
```

Output:

```
PS C:\Users\Chaitanya Barge\Desktop\Git Folder> & "C:\Users\Chaitanya Barge\AppData\Local\Programs\Python\Python38\python.exe" C:\Users\Chaitanya Barge\Desktop\Git Folder\array.py
a) 1D array of 10 random numbers between 0 and 1:
[0.6686187  0.27370409 0.1625371  0.85470736 0.12922679 0.18829113
 0.6243221  0.75516397 0.77410267 0.97129953]

b) 3x3 matrix of random numbers (standard normal distribution):
[[-0.83517716  2.01750682  1.04303293]
 [ 0.68213766  0.55849849  0.7174404 ]
 [-0.3515789  -0.56850611  0.25113217]]
```

Q.6

Ans

Program Code:

```
#Q.6
import numpy as np
v1 = np.array([2, 4, 6, 8])
v2 = np.array([1, 3, 5, 7])
addition = v1 + v2
print("Addition:")
print(addition)
subtraction = v1 - v2
print("\nSubtraction:")
print(subtraction)
multiplication = v1 * v2
print("\nElement-wise Multiplication:")
print(multiplication)
dot_product = np.dot(v1, v2)
print("\nDot Product:")
print(dot_product)
```

Output:

```
PS C:\Users\Chaitanya Barge\Desktop\Git Folder> & "C:\Users\Chaitanya Barge\AppData\Local\Programs\Python\Python38-32\python.exe" C:\Users\Chaitanya Barge\Desktop\Git Folder\array.py"
Addition:
[ 3  7 11 15]

Subtraction:
[1 1 1 1]

Element-wise Multiplication:
[ 2 12 30 56]

Dot Product:
100
```

Q.7

Ans

Program Code:

```
#Q.7
import numpy as np
A = np.array([
    [1, 2, 3],
    [4, 5, 6],
    [7, 8, 9]])
B = np.array([
    [9, 8, 7],
    [6, 5, 4],
    [3, 2, 1]])
addition = A + B
print("Matrix Addition:")
print(addition)
multiplication = A @ B
print("\nMatrix Multiplication:")
print(multiplication)
elementwise = A * B
print("\nElement-wise Multiplication:")
print(elementwise)
```

Output:

```
PS C:\Users\Chaitanya Barge\Desktop\Git Folder> & "C:\Users\Chaitanya Barge\AppData\Local\Programs\Python\Python313\python.exe" C:\Users\Chaitanya Barge\Desktop\Git Folder\array.py
Matrix Addition:
[[10 10 10]
 [10 10 10]
 [10 10 10]]

Matrix Multiplication:
[[ 30  24  18]
 [ 84  69  54]
 [138 114  90]]

Element-wise Multiplication:
[[ 9 16 21]
 [24 25 24]
 [21 16  9]]
```

Q.8

Ans

Program Code:

```
#Q.8
import numpy as np
array = np.random.randint(1, 101, (4, 4))
print("4x4 Random Integer Matrix:")
print(array)
print("\nShape:", array.shape)
print("Dimension (ndim):", array.ndim)
print("Total number of elements (size):", array.size)
print("Data type:", array.dtype)
print("Minimum value:", array.min())
print("Maximum value:", array.max())
```

Output:

```

PS C:\Users\Chaitanya Barge\Desktop\Git Folder> & "C:\Users\Chaitanya Barge\AppData\Local\Programs\Python\Python
lder/array.py"
4x4 Random Integer Matrix:
[[70 10 10 72]
 [22 60 95 18]
 [21 70 40 13]
 [52 9 60 19]]

Shape: (4, 4)
Dimension (ndim): 2
Total number of elements (size): 16
Data type: int32
Minimum value: 9
Maximum value: 95

```

Q.9

Ans

Program Code:

```

#Q.9
import numpy as np
array = np.random.randint(1, 51, 20)
matrix = array.reshape(4, 5)
print("4x5 Matrix:")
print(matrix)
print("\nSum:", matrix.sum())
print("Mean:", matrix.mean())
print("Standard Deviation:", matrix.std())
print("\nMaximum value in each row:")
print(matrix.max(axis=1))

```

Output:

```

PS C:\Users\Chaitanya Barge\Desktop\Git Folder> & "C:\Users\Chaitanya Barge\AppData\Local\Programs\Python
lder/array.py"
4x5 Matrix:
[[40 25 34 2 49]
 [27 26 16 17 34]
 [29 38 50 11 30]
 [3 10 30 2 24]]

Sum: 497
Mean: 24.85
Standard Deviation: 14.029522443761227

Maximum value in each row:
[49 34 50 30]

```

Q.10

Ans

Program Code:

```
#Q.10
import numpy as np
try:
    n = int(input("Enter how many random numbers you want to generate: "))
    if n <= 0:
        print("Please enter a positive number.")
    else:
        array = np.random.randint(10, 101, n)
        print("\nGenerated Array:")
        print(array)
        mean_val = np.mean(array)
        median_val = np.median(array)
        std_val = np.std(array)
        min_val = np.min(array)
        max_val = np.max(array)
        print("\nStatistics:")
        print("Mean:", mean_val)
        print("Median:", median_val)
        print("Standard Deviation:", std_val)
        print("Minimum:", min_val)
        print("Maximum:", max_val)
        print("\nReshaping into 2D array (if possible):")
        rows = int(np.sqrt(n))
        cols = n // rows
        if rows * cols == n:
            reshaped = array.reshape(rows, cols)
            print(reshaped)
            print("\nRow-wise Sum:")
            print(reshaped.sum(axis=1))
        else:
            print("Cannot reshape perfectly into a 2D matrix.")
except ValueError:
    print("Invalid input! Please enter a valid integer.")
```

Output:

```
PS C:\Users\Chaitanya Barge\Desktop\Git Folder> & "C:\Users\Chaitanya Barge\AppData\Local\Program  
lde/interpreter/array.py"
```

```
Enter how many random numbers you want to generate: 10
```

```
Generated Array:
```

```
[90 86 58 52 59 78 19 18 15 72]
```

```
Statistics:
```

```
Mean: 54.7
```

```
Median: 58.5
```

```
Standard Deviation: 27.022398117117582
```

```
Minimum: 15
```

```
Maximum: 90
```

```
Reshaping into 2D array (if possible):
```

```
Cannot reshape perfectly into a 2D matrix.
```